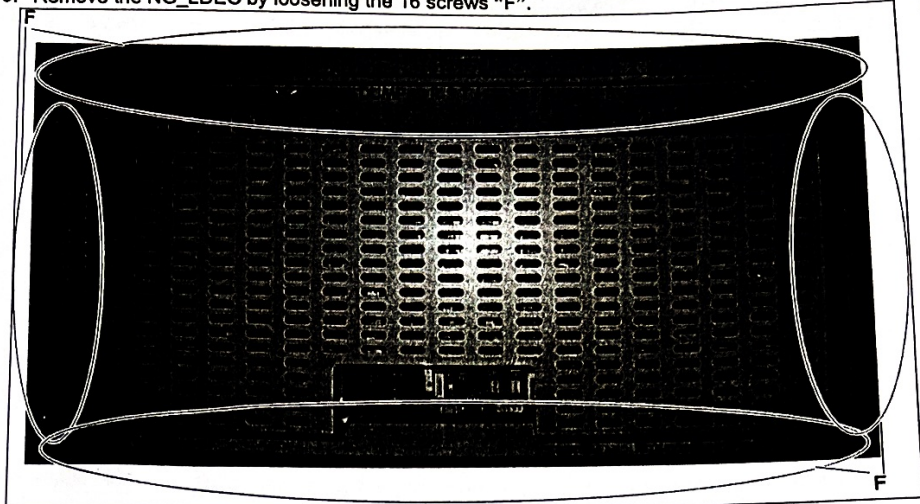
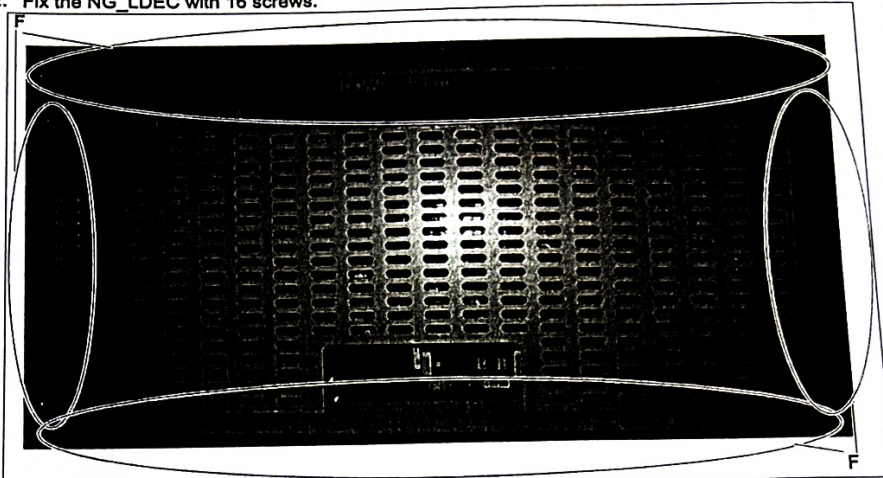


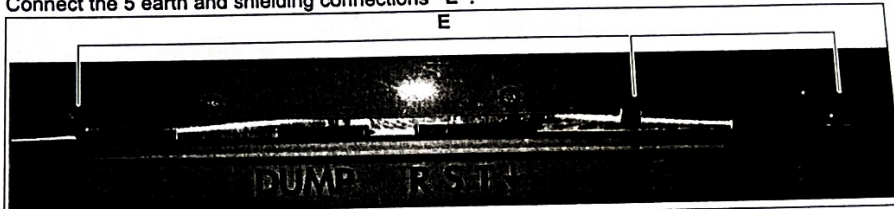
6. Remove the NG_LDEC by loosening the 16 screws "F".



1. Fit the new NG_LDEC into its position.
2. Fix the NG_LDEC with 16 screws.



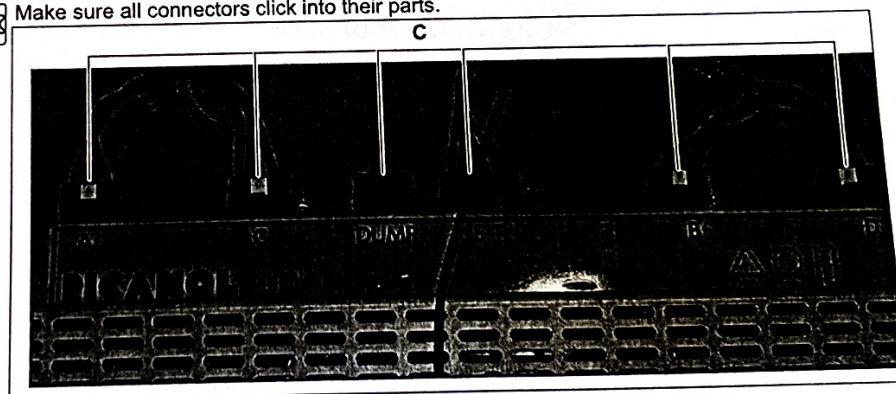
3. Connect the 5 earth and shielding connections "E".



4. Connect the connectors of the NG_LDEC: -A+; +C-; Dump; RST; -B+; +D-.



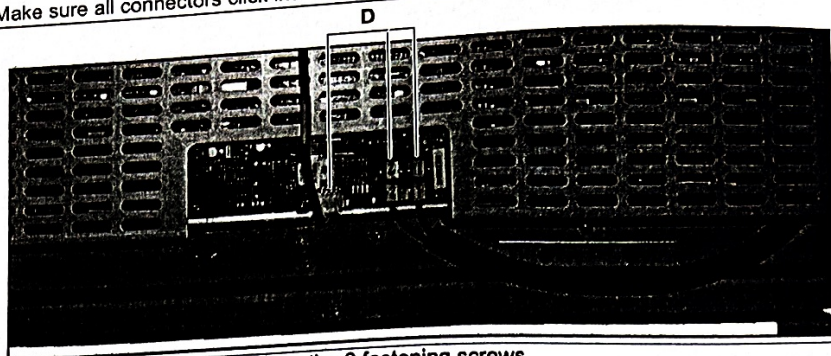
Make sure all connectors click into their parts.



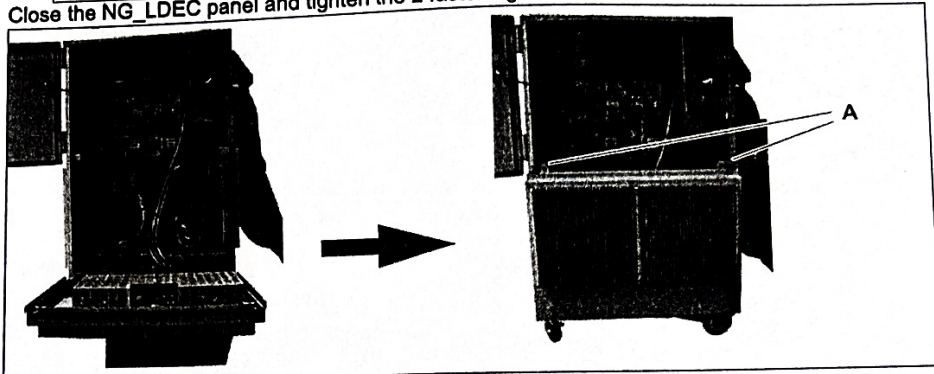
5. Connect the connectors "D" of the NG_LDEC: DD (P2); optical 2; optical 1.



Make sure all connectors click into their parts.



6. Close the NG_LDEC panel and tighten the 2 fastening screws.



7. Close the main control box.

NG_LDEC > OBMM110 print

Name	Output Board Main Motor 110KW
Version	OBBM110-1: BE221528
General	The OBMM print is part of the <u>NG_LDEC</u> [43].
Function	This print: • redirects the power stages for 2 motor coils; • measures the power at the motor phases; • monitors the temperature of the modules; • protects against desaturation of the modules; • generates tension for the drivers.  In the NG_LDEC there are 2 identical OBMM prints. One print controls motor phases A and C (OBMMac), the other print drives the motor phases B and D (OBMMbd).
Signals	There are no LEDs on this print.
Fusing	There are no fuses on this print.

Illustration	Connector	Connections OBMMac	Connections OBMMbd
	OBMM-01	Phase A	Phase B
	OBMM-02	Phase C	Phase D
	OBMM-03	connection with CBMM	connection with CBMM
	OBMM-04	not used	
	OBMM-S1	directly connected to EBMMac	directly connected to EBMMbd
	OBMM-S2		
	OBMM-S3		
	OBMM-S4		

Removing the OBMM

This procedure describes how to remove OBMM from the NG_LDEC.

The illustrations refer to OBMMac, but the procedure also applies to OBMMbd.

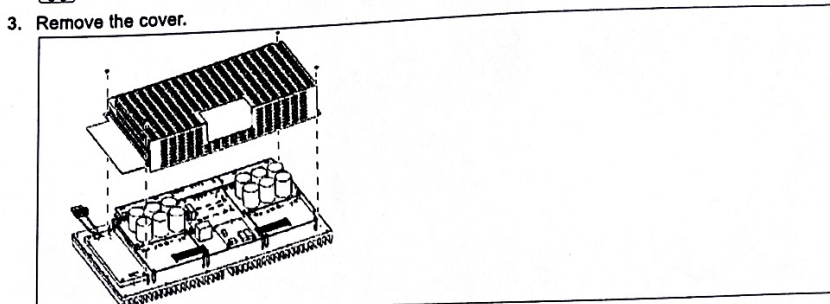
1. Remove the NG_LDEC panel from the control box. See [Removing the NG_LDEC](#) [49].

2. Store the NG_LDEC in a dust-free environment.

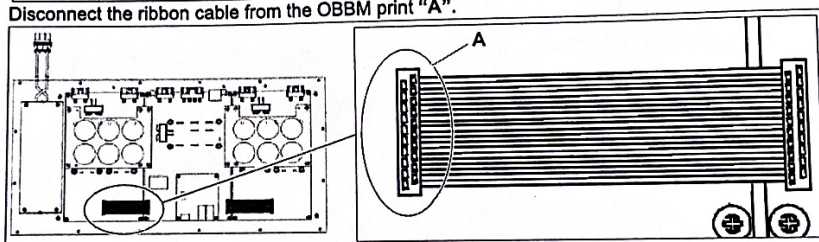
3. Remove the cover.



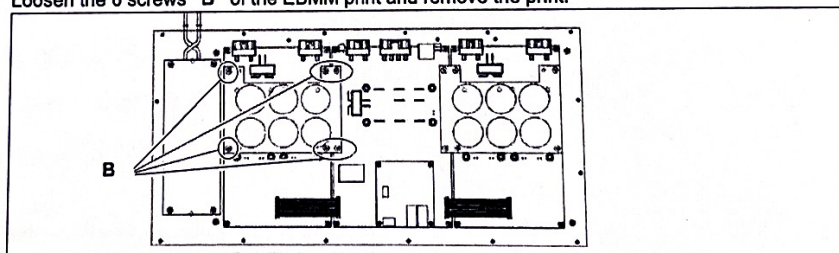
Dust between the power modules and the heatsink can damage the modules.



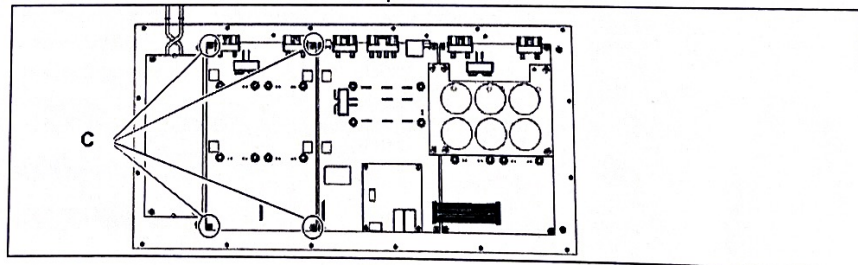
4. Disconnect the ribbon cable from the OBMM print "A".



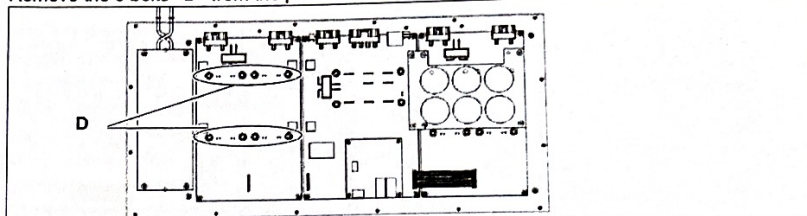
5. Loosen the 6 screws "B" of the EBMM print and remove the print.



6. Remove the 4 screws "C" from the OBMM print.



7. Remove the 8 bolts "D" from the power modules.



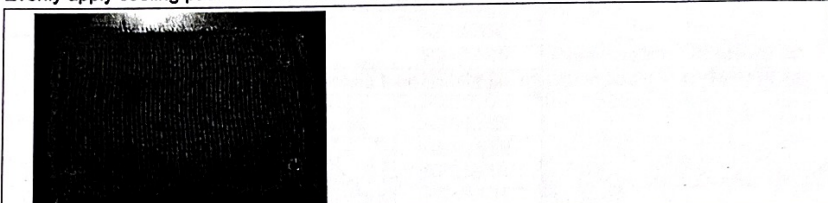
8. Remove the OBMM print.
9. Remove the cooling paste from the heatsink and from the OBMM power module.

Fitting the OBMM



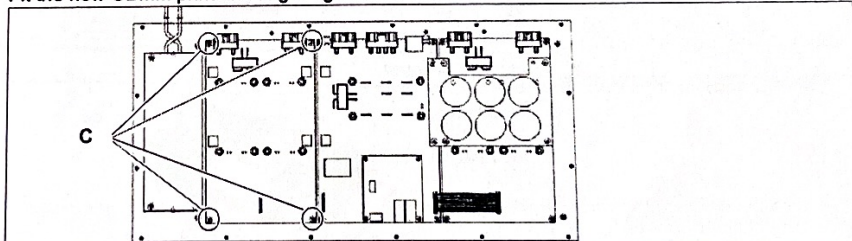
*This procedure describes how to fit the OBMM into the NG_LDEC.
The illustrations refer to OBMMac, but the procedure also applies to OBMMbd.*

1. Evenly apply cooling paste onto the heatsink.



Spread the cooling paste with a spatula.

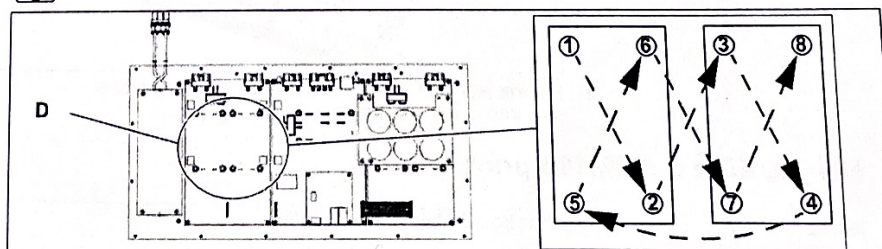
2. Fit the new OBMM print and finger-tighten the 4 screws "C".



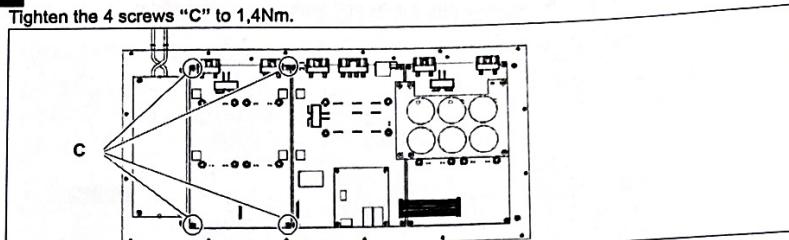
3. Tighten the 8 bolts "D" of the power modules.



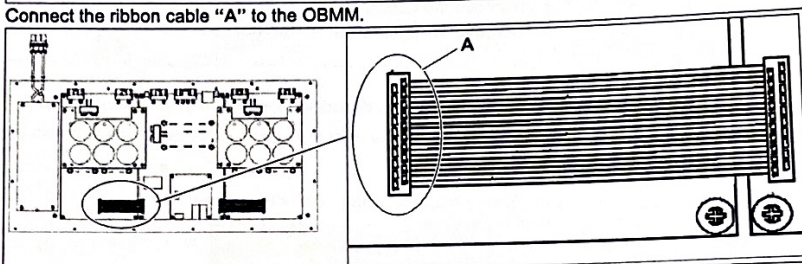
Tighten the bolts in diagonal order. Tighten first to 2Nm, afterwards to 4Nm.



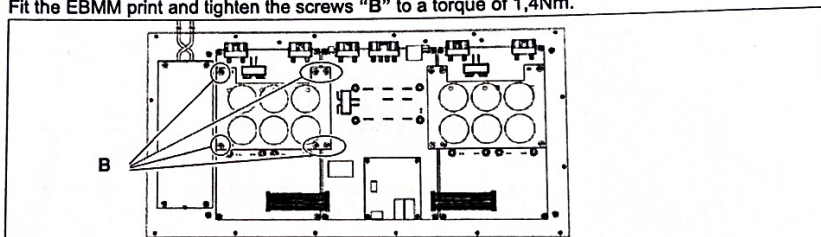
4. Tighten the 4 screws "C" to 1,4Nm.



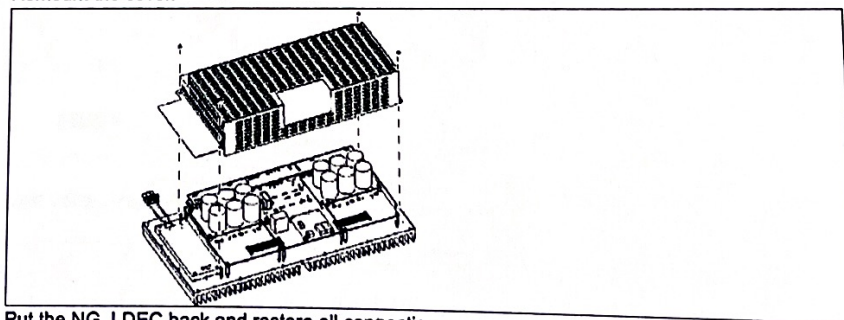
5. Connect the ribbon cable "A" to the OBMM.



6. Fit the EBMM print and tighten the screws "B" to a torque of 1,4Nm.



7. Remount the cover.



8. Put the NG_LDEC back and restore all connections.
See Fitting the NG_LDEC [51].

NG_LDEC > RBMM110 print

Name	Rectifier Board Main Motor 110KW
Version	RBMM110-1: BE221526
General	The RBMM110 print is part of the <u>NG_LDEC</u> [43].

Function

This print:

- converts the main voltage to direct current;
- controls and detects dump circuit;
- checks the tension;
- checks the current (differential);
- converts to low voltage;
- combines the protections;
- divides the control signals;
- offers the option to export the VDCL.

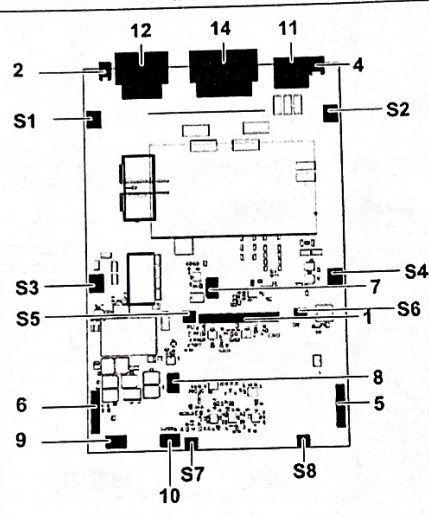
Signals

The signals are described in the NG_LDEC section.
See Signals [47].

Fusing

There are no fuses on this print.

Connections

Illustration	Connector	Connections
 The diagram shows a circuit board with various components labeled with numbers and letters. At the top, there are three large components labeled 12, 14, and 11. Below them are smaller components labeled 2, 4, S1, S2, S3, S4, S5, S6, S7, S8, and 10. The board is populated with numerous electronic components including resistors, capacitors, and integrated circuits.	RBMM-01	Directly plugged into the CBMM110 print
	RBMM-02	not used
	RBMM-04	
	RBMM-05	ribbon cable to OBMM110 bd
	RBMM-06	ribbon cable to OBMM110 ac
	RBMM-07	for Picanol service
	RBMM-08	
	RBMM-09	
	RBMM-10	
	RBMM-11	not used
	RBMM-12	dump resistor
	RBMM-14	power supply: R, S, T, PE
	RBMM-S1	directly connected to EBMM
	RBMM-S2	
	RBMM-S3	
	RBMM-S4	
	RBMM-S5	directly connected to CBMM
	RBMM-S6	
	RBMM-S7	
	RBMM-S8	

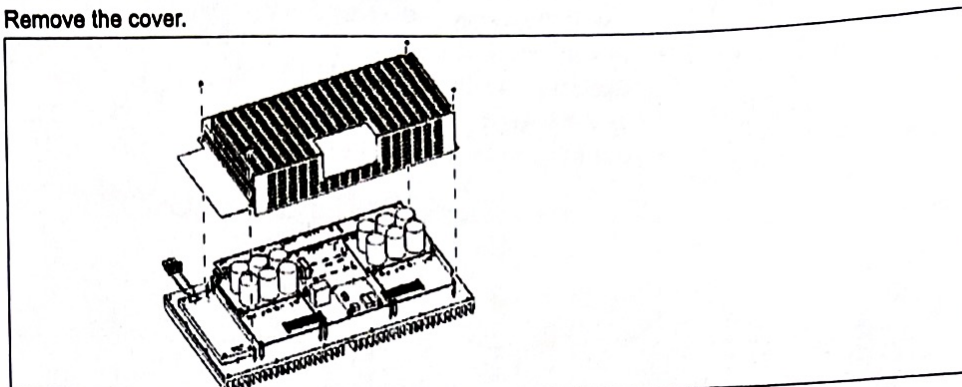
Removing the RBMM

1. Remove the NG_LDEC panel from the control box. See Removing the NG_LDEC [49].
2. Store the NG_LDEC in a dust-free environment.

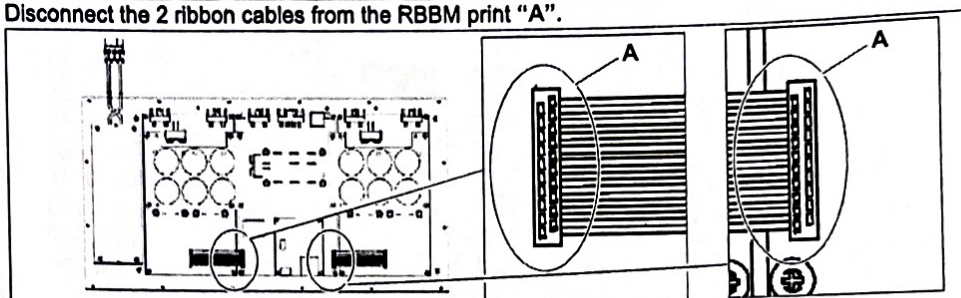


Dust between the power modules and the heatsink can damage the modules.

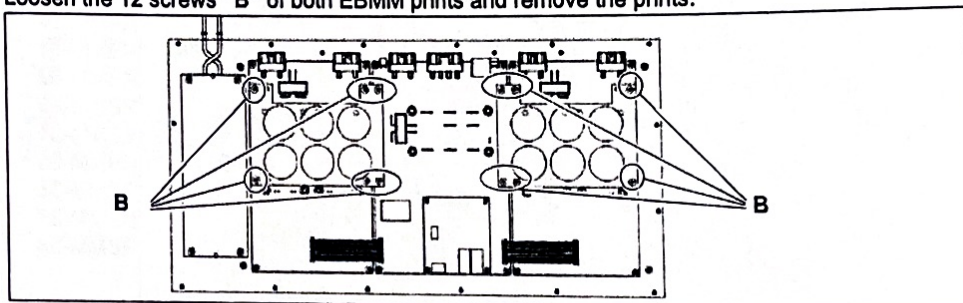
3. Remove the cover.



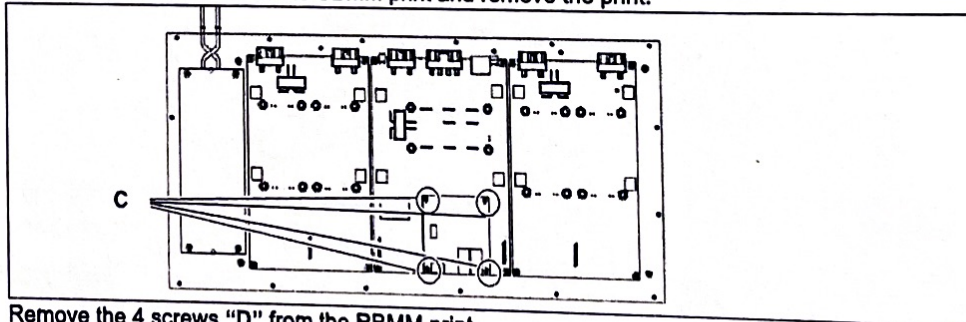
4. Disconnect the 2 ribbon cables from the RBMM print "A".



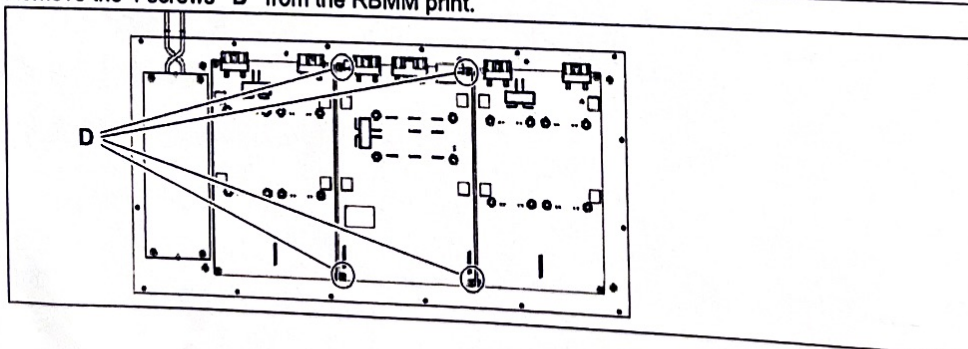
5. Loosen the 12 screws "B" of both EBMM prints and remove the prints.



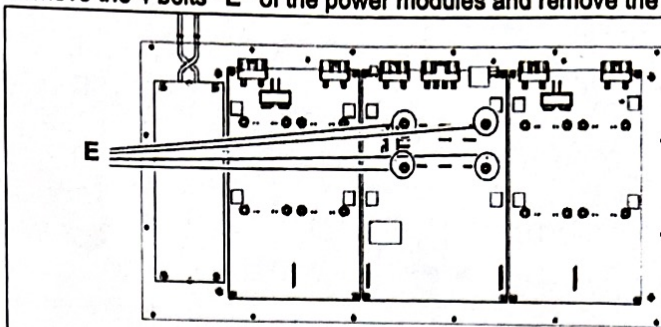
6. Loosen the 4 screws "C" of the CBMM print and remove the print.



7. Remove the 4 screws "D" from the RBMM print.



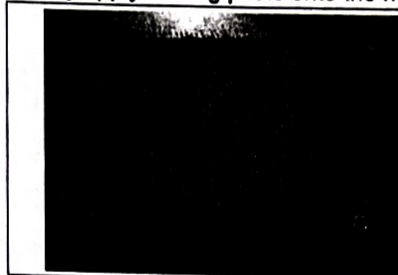
8. Remove the 4 bolts "E" of the power modules and remove the RBMM print.



9. Remove the cooling paste from the heatsink and from the RBMM power module.

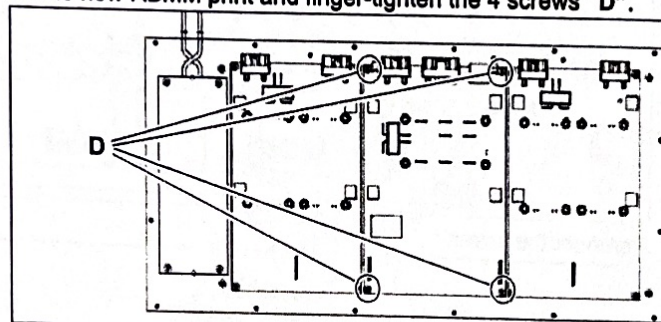
Fitting the RBMM

1. Evenly apply cooling paste onto the heatsink.



Spread the cooling paste with a spatula.

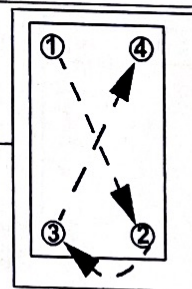
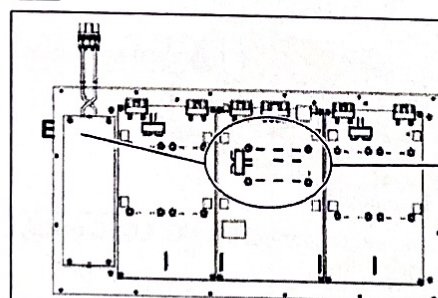
2. Fit the new RBMM print and finger-tighten the 4 screws "D".



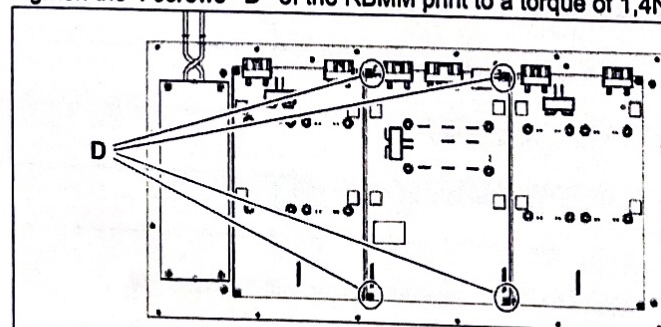
3. Tighten the 4 bolts "E" of the power modules.



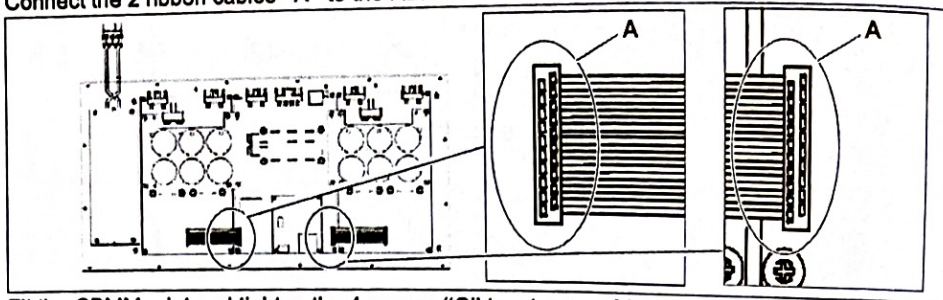
Tighten the bolts in diagonal order. Tighten first to 2Nm, afterwards to 4Nm.



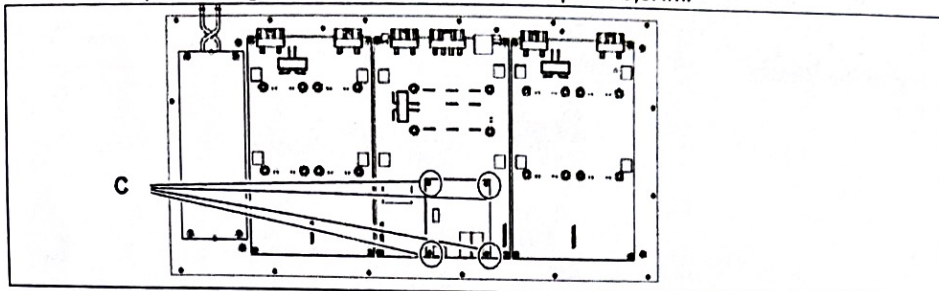
4. Tighten the 4 screws "D" of the RBMM print to a torque of 1,4Nm.



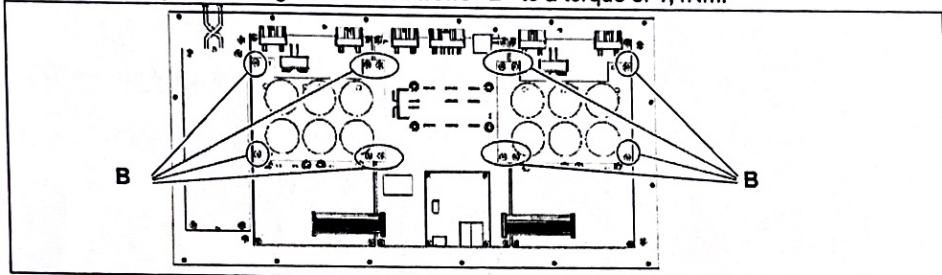
5. Connect the 2 ribbon cables "A" to the RBMM.



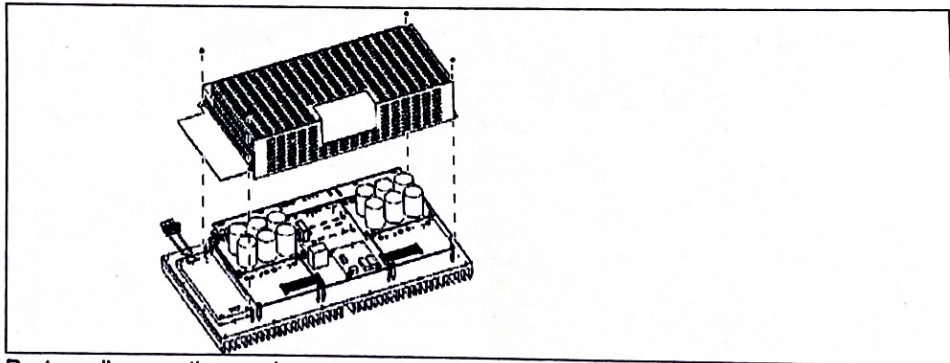
6. Fit the CBMM print and tighten the 4 screws "C" to a torque of 0,6Nm.



7. Fit the 2 EBMM prints and tighten the 12 screws "B" to a torque of 1,4Nm.



8. Remount the cover.



9. Restore all connections and reconnect the NG_LDEC panel.
See Fitting the NG_LDEC [51].

NG_LDEC > CBMM110 print

Name	Control Board Main Motor 110KW
Version	CBMM110-1: BE222346
General	The CBMM110 print is part of the <u>NG_LDEC</u> [43].
Function	This print: • provides optical communication with the machine; • controls the driver phases; • converts the measured values from analogue to digital; • checks and reacts to the protections; • converts to 3,3V low voltage; • shows the status by means of LEDs.

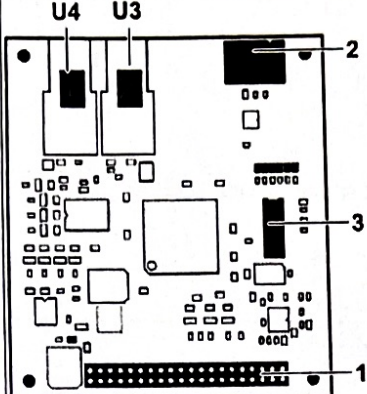
Signals

The signals are described in the NG_LDEC section.
See [Signals \[47\]](#) and [Error detection \[48\]](#).

Fusing

There are no fuses on this print.

Connections

Illustration	Connector	Connections
	RBMM-01	Directly plugged into the RBMM print
	RBMM-02	DD signal
	RBMM-03	not used
	U3	(receive) optical connection with the DDMP-U4 (2)
	U4	(transmit) optical connection with the DDMP-U2 (1)

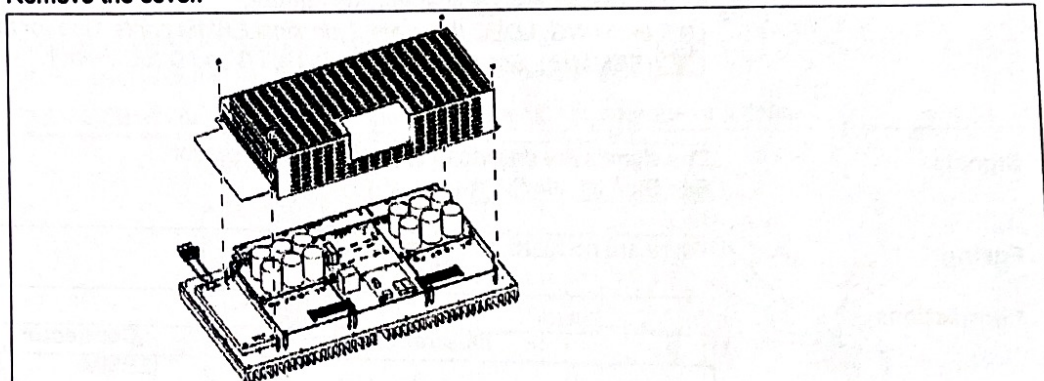
Removing the CBMM

1. Remove the NG_LDEC panel from the control box. See [Removing the NG_LDEC \[49\]](#).
2. Store the NG_LDEC in a dust-free environment.

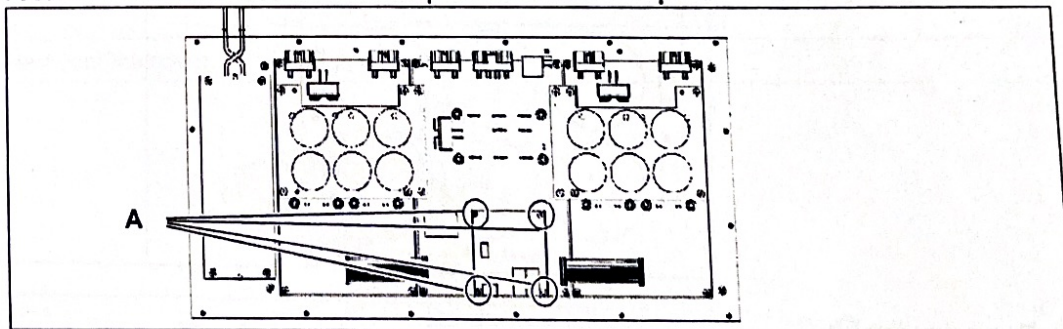


Dust between the power modules and the heatsink can damage the modules.

3. Remove the cover.

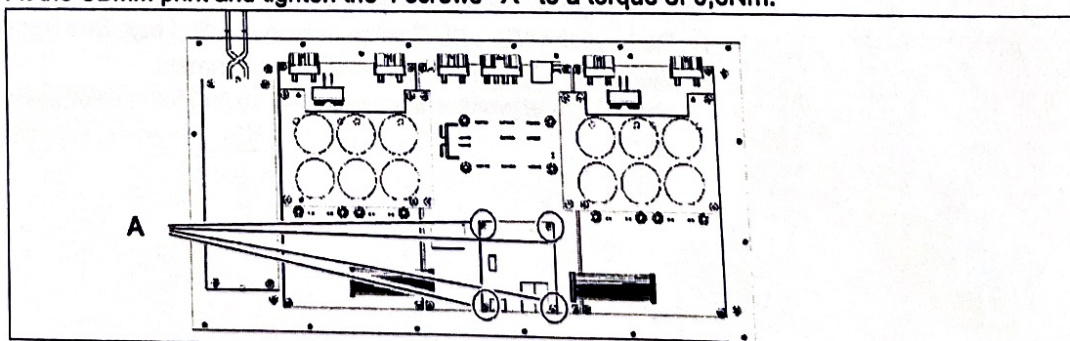


4. Loosen the 4 screws "A" of the CBMM print and remove the print.

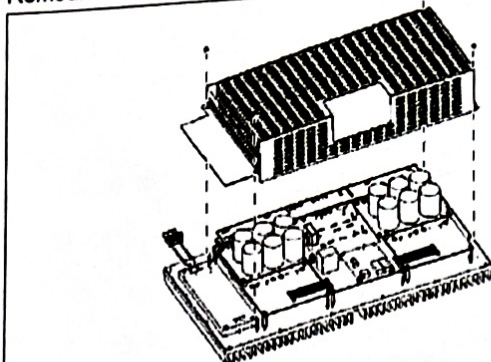


Fitting the CBMM

1. Fit the CBMM print and tighten the 4 screws "A" to a torque of 0,6Nm.



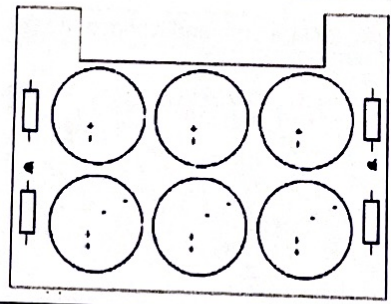
2. Remount the cover.



3. Restore all connections and reconnect the NG_LDEC panel.
See Fitting the NG_LDEC [51].

NG_LDEC > EBMM1020 print

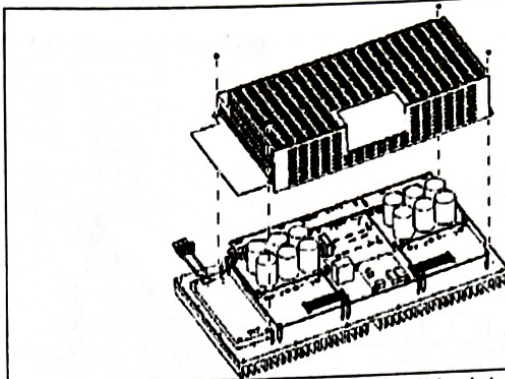
Name	ELCO Board Main Motor 1020 μ F.
Version	EBBM1020-1: BE222462
General	The EBBM020 print is part of the <u>NG_LDEC</u> [43].
Function	This print: • buffers and smoothens the peak capacities; • discharges energy after tension release.  In the NG_LDEC there are 2 identical EBMM prints. One for the motor phases A and C (EBMMac), one for the motor phases B and D (EBMMbd).
Signals	The signals are described in the NG_LDEC section. See <u>Signals</u> [47].
Fusing	There are no fuses on this print.
Connections	

Illustration	Connector	Connections
	EBMM	directly plugged into the 2 OBMM prints and the RBMM print

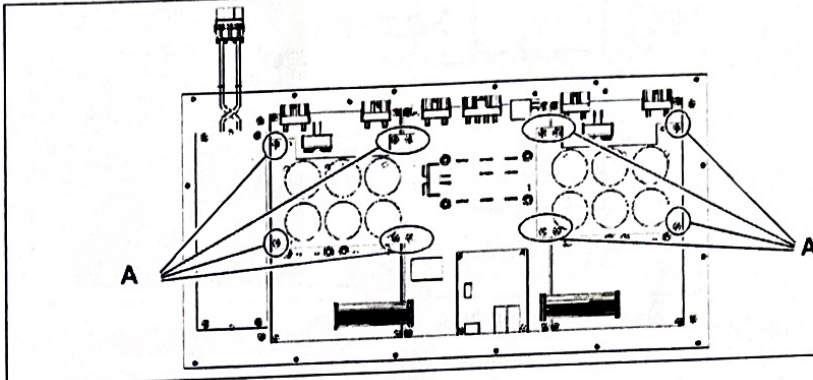
Removing the EBMM

-  Always replace both EBMM prints together.
If one print is defective, the other print will also be damaged.
1. Remove the NG_LDEC panel from the control box. See Removing the NG_LDEC [49].
 2. Store the NG_LDEC in a dust-free environment.
 Dust between the power modules and the heatsink can damage the modules.

3. Remove the cover.



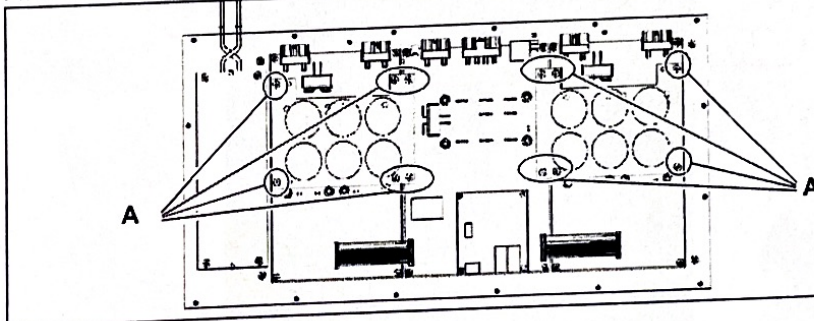
4. Loosen the 12 screws "A" of both EBMM prints and remove the prints.



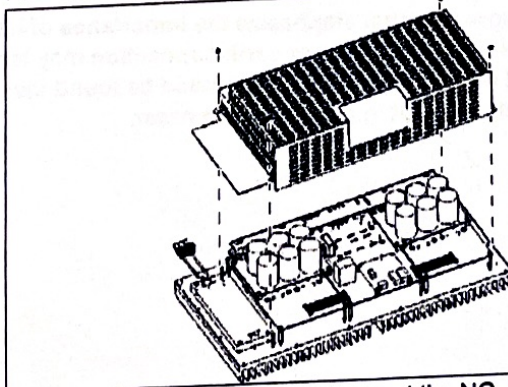
Fitting the EBMM

-  Always replace both EBMM prints together.
If one print is defective, the other print will also be damaged.

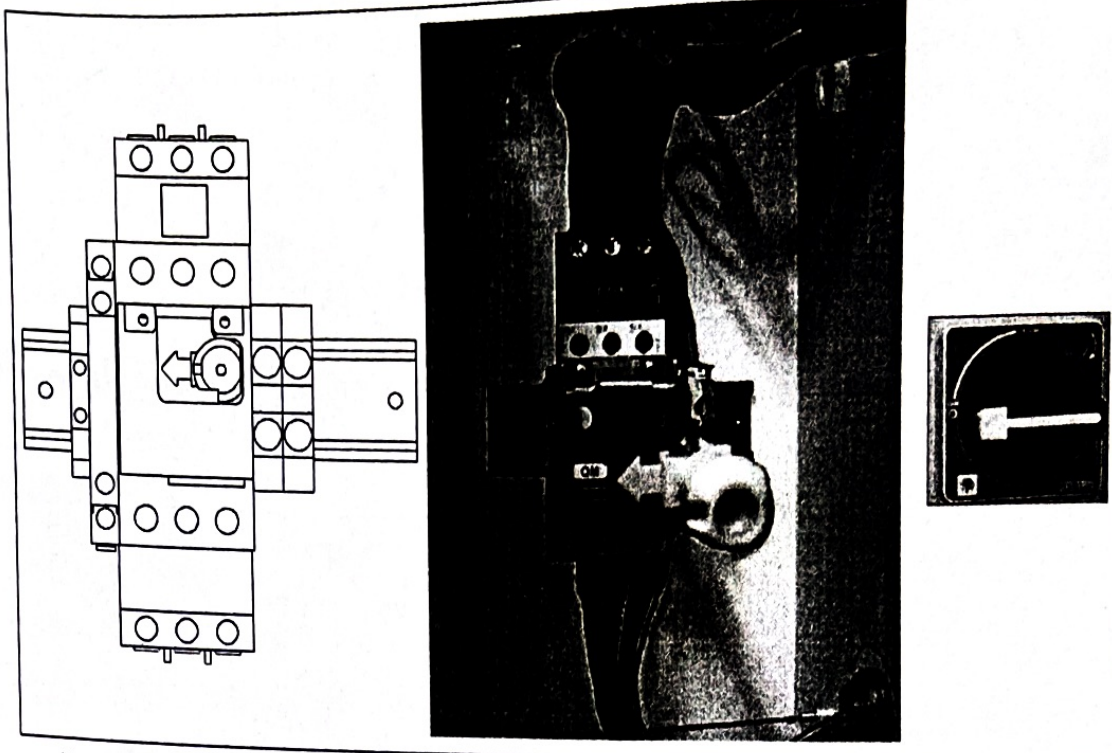
1. Fit the 2 EBMM prints and tighten the 12 screws "B" to a torque of 1,4Nm.



2. Remount the cover.

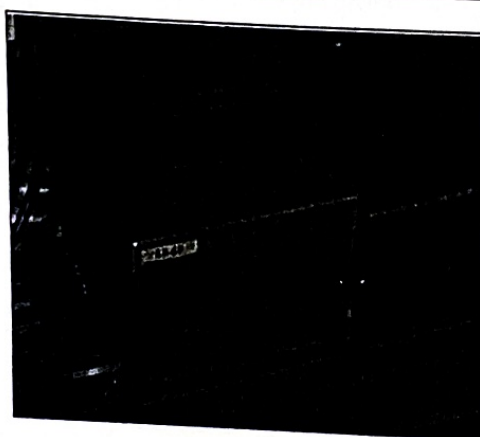


3. Restore all connections and reconnect the NG_LDEC panel.
See Fitting the NG_LDEC [51].



- Nominal current: 25 A
- Short circuit protection: 327 A $\pm 20\%$

FL: mains filter (only with CE machines)



Once more we must emphasize the importance of having a good earth connection in the weaving mill. A poor earth connection may lead to dangerous situations. Picanol rejects any liability if it should be found that the weaving mill's mains earth connection is not in good working order.

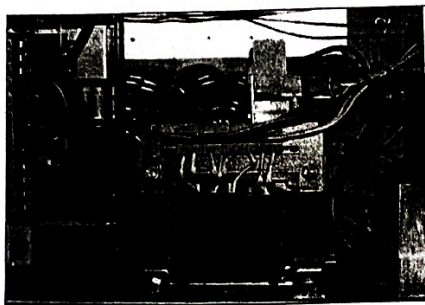
**LET'S GROW
TOGETHER** 
LR: coil

OptiMax-i



This coil serves to improve the power factor.

T-ILLUM: transformer for fabric lighting



Main control box (CB) > Low-voltage section

Opening the door to
the low-voltage section

See Side 1 [27].

Contents

- General [66]
- Components [66]
- MCB print [66]
- DDMP print [71]
- MIDB/R print [72]
- IBR print [73]
- UEB X and UEB Y [76]
- Board rack [77]
- Board rack > SBPO-print [77]
- Board Rack > ATB X, ATB Y and ATB Z print [78]

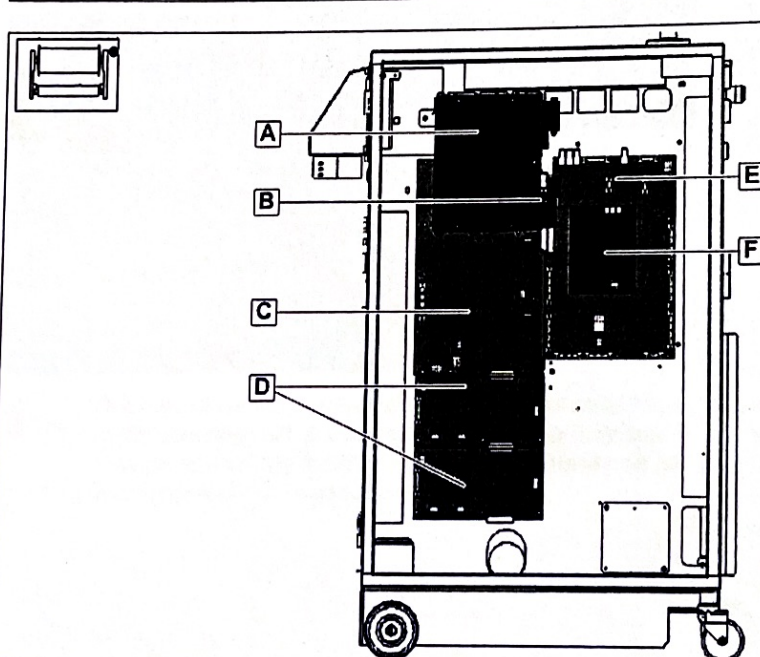
General

Electrostatic
discharges

The printed circuit boards contain integrated circuits that are very sensitive to electrostatic discharges:

- Put the printed circuit boards away in special boxes.
- Do not place the printed circuit boards on a metal surface, to avoid short-circuiting the batteries on the board.
- Do not touch the printed circuit boards unnecessarily and handle them with care to avoid permanent damage to the circuits.

Components



If the control box is not equipped with a board rack, the machine has a MIDB/R print [72] instead of a MIDSB print.

A. <u>Board rack</u> [77]	B. <u>MIDSB print</u> [72]
C. <u>IBR print</u> [73]	D. <u>UEB X and UEB Y</u> [76]
E. <u>MCB print</u> [66]	F. <u>DDMP print</u> [71]

MCB print

Name

Main Control Board

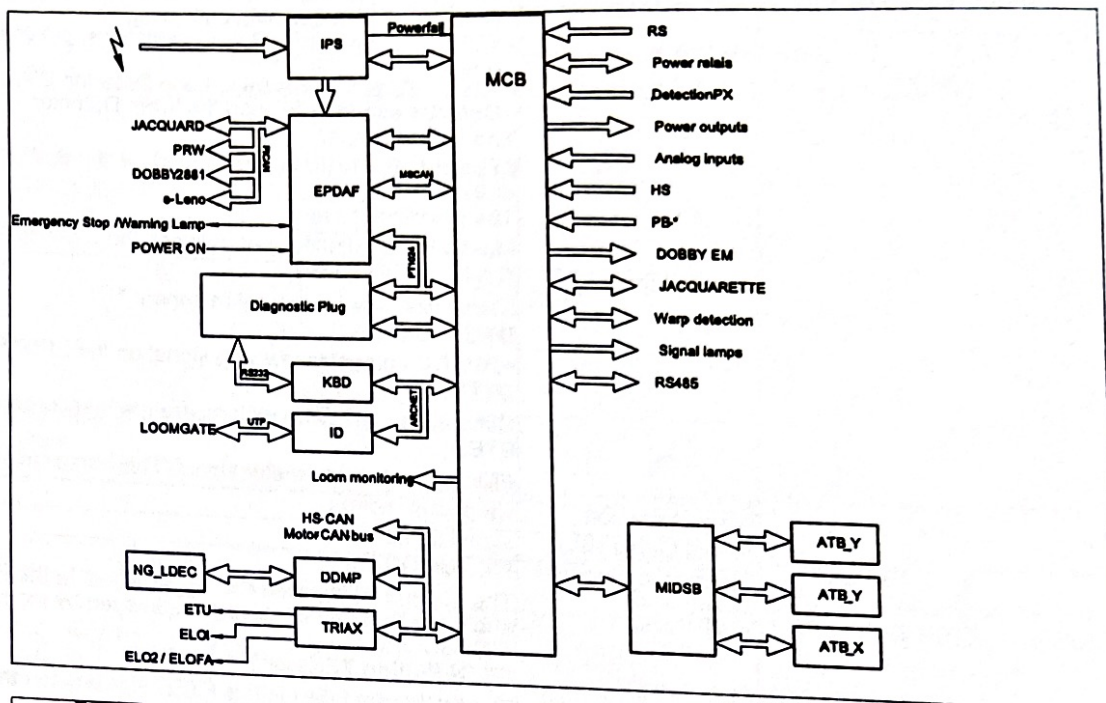
Version

MCB-7; BE222752

General

The MCB print contains a microprocessor using its own software. The software is loaded into flash EPROM's using the machine terminal.

Function

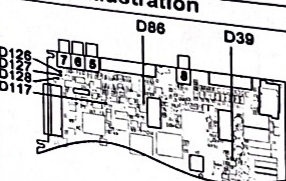


Function/ component	Description
Security	Together, the MCB and DDMP prints provide the protection logic.
Power fall	This signal is passed on to the microprocessor when mains power falls.
Loom monitoring	This parallel output contains the following lines: - start/stop; - warp stop; - selvedge stop; - filling stop; - pulse/revolution. - spare output. On the basis of these signals a weaving mill computer can calculate the efficiency of the machine.
RS (resolver main motor)	By means of two coils the resolver determines the exact angle position of the machine.
Hand Safety (HS)	The machine is equipped with a safety light grid.
EM DOBBY	This output is used to control the dobby of a type different from 2881.
Power relay	Print relays are used to connect the 48V AC for controlling the various motor switches: - K-MM: main motor switch; - K-SFTY: machine protection motor switch; - K-M-ASP: motor switch aspirator; - K-M-LVL-ON: leveling motor switches position ON; - K-M-LVL-OFF: leveling motor switches position OFF; - K-STRT-STP: start-stop motor switch.
Detection and PX	Detections - oil pressure detection DT-LU-PRESS; - oil filter contamination detection DT-LU-FL; - detection of the following supply voltages: +26V DC, 24V AC, 48V AC, +24V DCS and -12V DC. Proximity switches (PX) - PX-WP: detects whether the pickfinder is in weaving position; - PX-PFP: detects whether the pickfinder is in pickfinding position; - PX-REF: detects the reference-zone of the sley.

Function/ component	Description
Power outputs	On the MCB print 3 voltages are derived from the +26V DCS - +26V DCL1: power supply to the warp tension detection and calibration and power supply to the diagnostic plug; - +26V DCL2: power supply to detections, proximity switches and control buttons; - +26V WD: power supply to Leno Detector, Warp Detector, Waste Yarn Detector and the LEDs on the Warp Detector. The safety brake At power off, the brake is engaged mechanically and it must be energized in order to disengage. The pickfinder valve VPF Moves the pickfinder gear between the weaving and pickfinder positions.
Analog Inputs	DT-WT* (* = 1 and 2) Signal from the warp tension sensor * DTI26VDC +26V DC current measuring signal on the EPDAF print DTTHLU Circulating lubrication oil temperature measurement FTEN Filling yarn tension meter signal. This comes in via the diagnostic plug.
Warp Detection	warp stop motion
Signal Lights (SL)	signal lights
Diagnostic plug (X-DGN)	PICASCOOP The MCB print supplies the trigger signal to the Picascope while the trigger angle position can be advanced/delayed by means of a switch located on the PICASCOPE. FTEN (Filling TENSION) The MCB print takes in this FTEN signal when the filling tension meter is connected to the diagnostic plug. PT1024: PICAN communication pulse train The pulse train contains the angle position of the machine. RTX-CAN-2: bidirectional CAN communication
Serial communication	RS485 ARCNET The MCB print uses this serial line to communicate with: - the Keyboard Display; or - Interactive Display. HS-CAN (Motor CAN bus) - DDMP <=> NG LDEC: Up to the DDMP print, HS-CAN is an electrical signal. On the DDMP print HS-CAN is converted into an optical signal for the NG_LDEC module. The angle position sent from the DDMP print to the NG_LDEC module is also an optical signal. - TRIAX: Up to the TRIAX and T_TRIAX print, HS-CAN is an optical signal. The angle position sent from the MCB print to the TRIAX and T_TRIAX print is an optical signal too. MS-CAN (Medium Speed CAN bus) The MS-CAN bus is now only used as a component of the PICAN bus.
PICAN	The PICAN bus is composed on the EPDAF print, using the following signals: - PT1024: PICAN communication pulse train; - MS-CAN: CAN communication; - The +26V DC power supply and PGD ground signals.
The push-buttons (PB-*)	The control buttons (PB-*) (* = SMF, SMR, PFF and PFR) All control buttons that may cause the machine to move, are double contact buttons: - PB-* -1 sends a signal to the microprocessor; - PB-* -2 is incorporated into the hardware safety system. The emergency stops (PB-ES-*) (* = CB, LHS, RHS and RRHS) These push-buttons are equipped with double contacts and both are taken in (separately) through the EPDAF print.

Function/ component	Description
Connections of signals via the EPDAF print	Warning lights (WL) The MCB print controls the warning lights via the EPDAF print. If one of the warning lights is faulty, a message appears: "check warning lights" on the terminal. Detection POWER ON button position (DT-PWRON) DD* (Drive Disable) bidirectional signal (* = 1; 2 and NG_LDEC) - DD1 (T_TRIAX print); - DD2 (TRIAx print); - DD-NG_LDEC (NG_LDEC module).
The control and check of ...	• IBR print; • UEB_X print; • UEB_Y print; • MIDB/R print or MIDSB print; • ATB_X print; • ATB_Y print; • ATB_Z print.
The communication with the JACQUARETTE	

Signals

Illustration	LED	Color	Name	The LED...
	D39	green	—	
	D86	green	Reset	comes on if +12 V DC is present on the print. is lit if the +5,2VDC on this print is more than 5 VDC.
	D117		Watchdog	If the +5,2 VDC is less than 5VDC, the RESET line is low. The microprocessor halts and will not start again until the +5,2 VDC once more exceeds 5 VDC.
	D126	yellow	HS-CAN-IN	- flashes if the microprocessor and the software are operating. - is out if the software 'hangs'. As a result there is no communication with the machine-terminal.
	D127		HS-CAN-OUT	- lights up weakly if the optical communication (5) between the MCB and TRIAX prints is operating properly. - lights up brightly or is off if a communication error has occurred.
	D128		RS-OUT	- lights up weakly if the optical communication (6) between the MCB and TRIAX prints is operating properly. - lights up brightly or is off if a communication error has occurred.
There are no fuses on the print.				

Fusing